Build a parcel holder

Name: ____________________

Date: ____________________ **Period:** ____________________

Today I can...

Build two versions of a holder and use your tests to choose one.

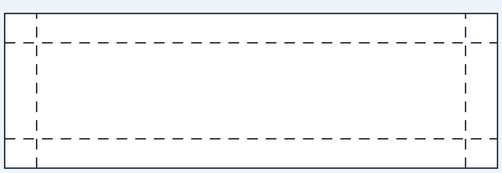
Words to know

Cargo: The thing your robot carries.

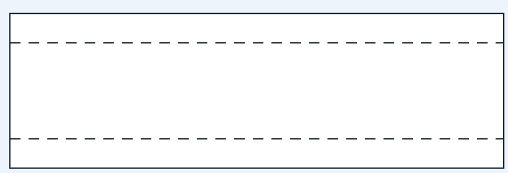
Cradle: A holder that keeps the parcel in place.

Fair test: Change one thing and keep the rest the same.

LOOK AT THE IDEA



A: four low walls



B: open short ends

Gather your materials

- Checked base with teacher-approved fixed mounting points
- Thin cardboard: two rectangles about 12 cm by 8 cm
- Paper parcel about 4 cm by 3 cm by 2 cm
- Ruler, pencil, scissors and removable tape

Build two parcel holders

Build with the power off. Tick each step when it is done.

1. Fold a small paper parcel about 4 cm long, 3 cm wide and 2 cm high. Tape it closed. Use this same light parcel for every test. Do not add coins or other weights.
2. Measure the clear space your teacher chose on the base. A 12 by 8 cm card is a starting size. Make it smaller if it would hang over a wheel or cover a control.
3. For holder A, draw a line 1 cm inside each edge of one rectangle. Cut out the four corner squares. Fold the four sides up along the lines and tape the corners. You have made a low tray.
4. For holder B, use the second rectangle. Fold its two long edges up by 1 cm and keep both short ends open. Put two folded paper stops near the short ends to stop the parcel sliding out.

Sketch the setup. Label the parts you will check.

Finish the setup

5. Turn the robot off. Ask the teacher to show the mounting points. Fix holder A to those fixed points with the approved reusable connectors or paper tape tabs. Do not tape over the motor, sensor or buttons.

6. Put the parcel in holder A. Raise the wheels and check for rubbing. Try the same short drive from lesson 2 three times and record whether the parcel stays on.

7. Turn the power off. Swap to holder B. Use the same parcel, code, floor and start line for three more runs.

8. Choose the better holder using your notes. Change only one feature if neither works well. Keep the selected holder for later lessons.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Show where your robot starts and how you will stop it.

Code it. Predict. Try it.

Make the program

1. Use lesson 2’s checked program.

2. Low speed, 0.8 seconds forward, then stop and end.

3. Keep the same settings for both holders.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.